Sequences and series

1 A sequence is a list of terms, and a series is the sum of the terms.

2 6

3 196

4 a_5

5

a Finite **b** Infinite **c** Finite **d** Infinite

6 Finite

7

a Infinite **b** Finite

8

a $0, 3, 6, 9, 12$ **b** $-2, -4, -8, -16, -32$

c $4, 8, 16, 32, 64$ **d** $-6, 24, -72, 192, -480$

e $1, 0.7071, 0.5774, 0.5, 0.4472$ **f** $2, 2.25, 2.3704, 2.4414, 2.4883$

9

a $\frac{5}{4}, \frac{6}{5}, \frac{7}{6}, \frac{8}{7}$ **b** $\frac{2}{3}, 1, 1, \frac{19}{21}$ **c** $-1, -\frac{1}{4}, 0, \frac{1}{16}$ **d** $0, -1, 2, -3$

10

a $4, 3, 4, 11$ **b** $3, 5, 8, 12$ **c** $-2, -2, 0, 8$ **d** $3, 5, 15, 75$

e $3, 2, 0, -8$

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Sigma notation

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a $\sum_{k=1}^{\infty} 4k$

b $\sum_{k=1}^{6} (-3)^k$

c $\sum_{k=1}^{5} \frac{1}{k^3}$

d $\sum_{k=1}^{7} \frac{1}{3k}$

e $\sum_{k=1}^{11} \frac{2}{5+k}$

f $\sum_{k=1}^{11} \left(-\frac{1}{2}\right)^{k-1}$

12

a 40

b $\frac{47}{60}$

c $\frac{19}{20}$

d 15

e 14

f 1815

g 368

h -16

i $\frac{63}{2}$

j 38

13 20

14 -10

15 30

16 -8

17

a -6

b -8

18

a -54

b 65

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19

a

$$\sum_{i=1}^{7}\left(i+i^3\right)=2+10+30+68+130+222+350$$

$$\sum_{i=1}^{7} i+\sum_{i=1}^{7} i^3=(1+2+3+4+5+6+7)+(1+8+27+64+125+216+343)$$

$$=2+10+30+68+130+222+350$$

b Yes

c Yes

Applications

20 $m_1=2000,\ m_n=(0.76)\,m_{n-1}$

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a $\$1, \$3, \$7, \$15$ **b** $\$38, \$43, \$48, \$53$ **c** Day 6

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